

Anant Rastogi

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SUMMARY

Firmware Engineer with 2+ years of experience developing embedded systems using Zephyr RTOS and ARM Cortex-M microcontrollers. Proven expertise in firmware development, automated testing frameworks, and CI/CD pipelines. Strong background in C/C++, real-time systems, JTAG debugging, and cross-platform integration.

TECHNICAL SKILLS

Embedded Systems: C/C++, Zephyr RTOS, ARM Cortex-M, GPIO, PWM, ADC, DAC, I2C, UART, TCP/IP, UDP, JTAG

Development Tools: Git, GitHub Actions, CI/CD Pipelines, MCUEXpress, Visual Studio, Python, Linux/Unix, Oscilloscopes

PROFESSIONAL EXPERIENCE

Firmware Design Engineer | Privoro LLC, Phoenix, AZ

August 2024-Present

- Developed **5+ firmware features** and modules using **C on Zephyr RTOS** for ARM Cortex-M microcontrollers, contributing to major firmware releases
- Designed and maintained an automated testing environment using GitHub Actions **CI/CD pipelines** and automated firmware test fixtures with **Python automation** for firmware validation, reducing integration issues and saving around 8 hours of developer time per release
- Investigated and resolved **25+ firmware bugs** using **JTAG debuggers** and testing environment, improving system stability and optimizing performance benchmarks
- Collaborated with application and cloud teams (6+ engineers) to resolve cross-platform bugs and validate system integration before releases

GNSS Engineer Intern | Hemisphere GNSS Inc., Scottsdale, AZ

May 2023-December 2023

- Upgraded GNSS receiver **Firmware** in **C/C++ and ARM Assembly** to newer ARMCC6 and GCC compilers, ensuring modern compatibility and improved performance.
- Deployed **mutex locks** for data protection across **6 threads**, ensuring thread-safe operations and preventing data corruption in the critical code section
- Acquired foundational knowledge of GNSS positioning algorithms, encompassing core principles and methods for accurate user positioning using satellite data.

Full Stack Developer | 247ERICPOINTCOM, Cape Town, South Africa (remote)

January 2023-April 2023

- Collaborated with a **team of 6 developers** globally to build a math quiz website <https://mathemacontest.com/> for high school students, contributing to **20% of the database design** and **80% of front-end components**
- Created the website's interactive front-end using **PHP and JavaScript**, creating timed quiz functionality with automatic scoring and a database of **1000+** math problems
- Built the website's database architecture using **MySQL**, enabling efficient data storage for user management, quiz results, and student progress tracking

RELEVANT PROJECTS

Undergraduate Thesis: Digital Elevation Models (DEMs) of the Moon and Mars from Optical Images

October 2023 - May 2024

- Conducted in-depth analysis and benchmarking of existing Image-to-DEM techniques, focusing on key scientific exploration sites on the Moon and Mars, to identify the highest-performing methods.
- Utilized over **300 low-resolution optical images** to successfully create accurate **3-D DEMs** of the unmapped terrains of the Moon and Mars, contributing significantly to the understanding of their topographies.

Autonomous Vehicle | C, FRDM-KL46Z board, HC-SR04 distance sensor, TCS34725 RGB Color Sensor

December 2023

- Created an autonomous vehicle prototype using **C and FreeRTOS**, achieving a **95% successful navigation rate**
- Orchestrated a closed-loop speed regulation architecture, leveraging ultrasonic sensors and UART communication, which improved speed control precision to within +/-2% and obstacle detection in 99% of trials.
- Applied **PWM control** for motor regulation and combined RGB and line sensors for accurate lane keeping with **±1cm** positioning accuracy

EXTRACURRICULAR EXPERIENCE

Undergraduate Teaching Assistant | Ira A. Fulton Schools of Engineering, ASU

August 2021-May 2023, January 2024-Present

- Courses Taught: Principles of Programming, Computer Organization and Assembly Language Programming, Grand Challenges for Engineering, Embedded Microprocessor Systems, Circuits.

Treasurer | Bridges International, Tempe, AZ

January 2022- Present

Lead Tutor | University Academic Success Programs, ASU

October 2022-May 2023

EDUCATION

BSE Computer Systems

August 2020-May 2024

Ira A. Fulton Schools of Engineering, Arizona State University, Tempe, Arizona

Cumulative GPA: 4.00

Honors and Awards: Barrett, The Honors College, Grand Challenges Scholar, Dean's List (Fall 2020 – Present), SUN Award, New American University Scholarship (NAMU), 17th Annual WiCS Programming Competition (1st Position)